

**LAB EXPERIMENT 5**  
**IMAGE CLASSIFICATION WITH NEURAL NETWORKS ON**  
**JETSON ORIN NANO**

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**Acknowledgment: I acknowledge all of the work (including figures and codes) belongs to me and/or persons who are referenced.**

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## LAB 5 SUMMARY

In this lab, we explored the end to end process of building an image classification model using neural networks. The idea was not just to train a model and see numbers go up, it was to understand the entire flow: from how datasets are created and preprocessed, to how a model learns, evaluates, and makes predictions. We worked with the Fashion MNIST dataset, which contains grayscale images of clothing items across 10 different categories, and our job was to guide this dataset through the full deep learning pipeline.

We started by learning about the importance of datasets in deep learning. Building a dataset isn't just about grabbing images online, it's a labor-intensive, manual process involving selection, trimming, resizing, and labeling. The reason this is such a big deal is because a neural network only learns what you feed it, if the dataset is off, the model's gonna be off too. In our case, we used a pre built dataset (Fashion MNIST), but we were made aware of how much work goes into preparing one.

Next, we stepped into the architecture of neural networks. The lab explained how each layer, especially the hidden layers, processes the inputs through weights and biases, and how backpropagation allows the model to learn by adjusting those weights after comparing predictions to the ground truth. This learning process was broken down in a way that was easy to follow, and I understood how each neuron contributes to the final prediction.

From there, we moved into the actual implementation. Using Python and TensorFlow (with Keras), we:

- Loaded and normalized the dataset (dividing by 255 to scale pixel values from 0–1)
- Visualized samples to get a feel for the data
- Built a neural network model with a Flatten layer and two Dense layers (one with 128 nodes using ReLU, and one output layer with 10 nodes using Softmax)
- Compiled it with the Adam optimizer and a sparse categorical crossentropy loss
- Trained the model for 10 epochs

The training phase gave us a model that hit ~90% accuracy on the training set. We then evaluated it on a test set, where accuracy dropped a little, a classic sign of overfitting, which the lab also discussed in detail. Overfitting means the model gets too used to the training data and performs worse on new, unseen data. To handle that, the dataset was split into training and testing sets from the beginning, which is a simple but effective way to validate the model.

Finally, we ran inference on test images and used prediction graphs to visualize which classes the model was most confident about. This helped us identify both correct and incorrect predictions and made it very clear how confidence scores can guide interpretation of results.

The preliminary questions tied everything together:

- We researched the CIFAR-100 dataset to compare it with Fashion MNIST
- We explained why dataset creation is so time-consuming (manual prep, labeling, etc.)
- We looked into how neural nets distinguish cats from dogs, showing how feature extraction works layer by layer
- And we discussed whether using training data for inference is a good idea, and understood why it can give high accuracy but still fail in real world generalization

Overall, this lab wasn't just about running code. It was about understanding what happens under the hood, step by step, and getting hands on with how deep learning actually works when training and deploying models on real devices.